Foundation models for EEG analysis

From raw EEG to robust insights

Julián David Pastrana-Cortés David Augusto Cárdenas-Peña Alvaro Ángel Orozco-Gutierrez

Universidad Tecnológica de Pereira

December 15, 2025

Outline

- **Motivation**
- **What is EEG?**
- **What is a Foundation Model?**
- **State of Art models**
- **Challenges**

Evolution of AI in Clinical EEG Analysis

1990s	2010s	2017+	2023+
Classical ML	**Deep Learning**	**Transfer Learning**	**Foundation Models**
SVM, RF	CNN, RNN	Pretrained	Self-supervised
Hand-crafted	End-to-end	Adaptation	Zero-shot
Single-center studies	Seizure detection	Cross-dataset transfer	Universal representations

From task-specific models to **general-purpose** representations that transfer across datasets and clinical applications.

Problem Statement

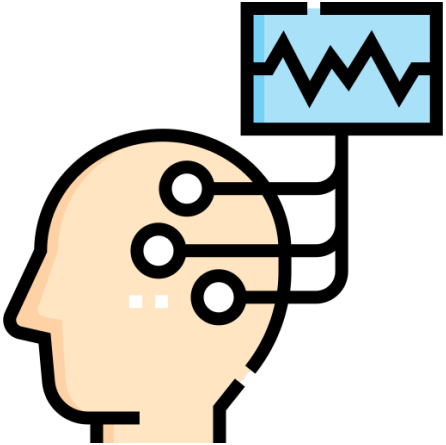
Approach	Complex Patterns	Low Label Requirement	Cross-Dataset Transfer	Zero-Shot Capability	Multimodal Integration
Classical ML (SVM, RF)	✗	✗	✗	✗	✗
CNN / RNN	✓	✗	✗	✗	✗
Transfer Learning	✓	✓	✓	✗	✗
Self-Supervised	✓	✓	✓	✗	✗
Foundation Model	✓	✓	✓	✓	✓

Foundation models take advantage of large-scale self-supervised pretraining, enabling strong generalization, zero-shot inference, and multimodal integration capabilities.

What is EEG?

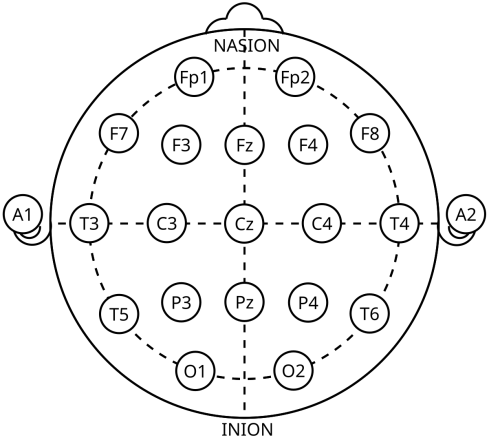
Electrical brain activity recorded from scalp electrodes. High temporal resolution, non-invasive measurement of neural dynamics.

Band	Range
Delta	0.5–4 Hz
Theta	4–8 Hz
Alpha	8–12 Hz
Beta	12–30 Hz
Gamma	$>$30 Hz



EEG Analysis Examples

- Supported Diagnosis
- Brain-Computer Interfaces
- Sleep Stage Classification
- Emotion Recognition
- Seizure Detection



Data preprocessing pipeline

Filtering

- Bandpass filter (1-50/80 Hz)
- Notch filter (50/60 Hz)

Artifact Removal

- Independent Component Analysis (ICA)
- Automatic algorithms
- Manual inspection

Segmentation

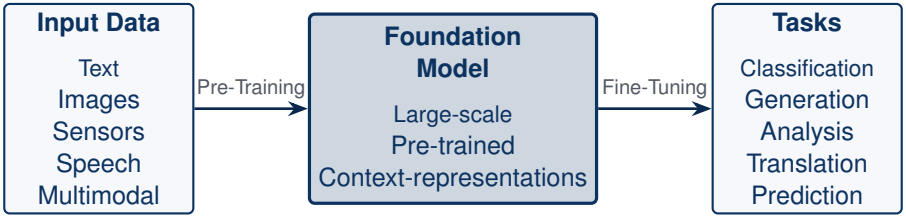
- Epoching based on events
- Fixed-length windows
- Overlap handling

Normalization

- Z-score normalization
- Min-max scaling
- Power normalization

Foundation Models

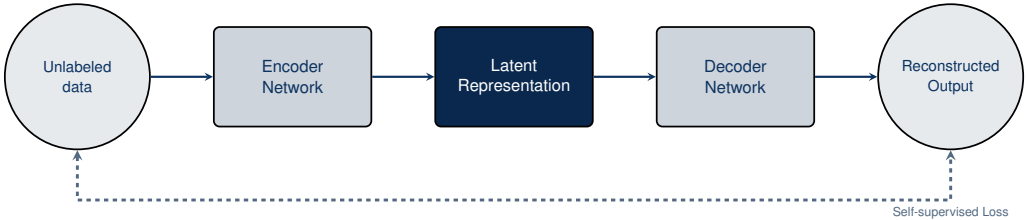
- Foundation models are large-scale machine learning models trained on vast amounts of data across various domains and modalities.
- They can be fine-tuned for specific downstream tasks with relatively small-labeled datasets



- A foundation model learns general representations from diverse data during pre-training.

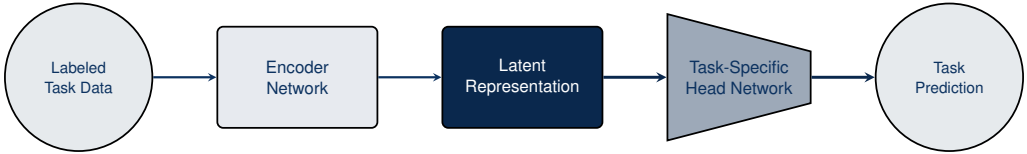
Breaking Down Foundation Models: Pre-training

The model is trained on a large and diverse dataset using self-supervised learning techniques. This allows the model to learn general features and patterns from the data.

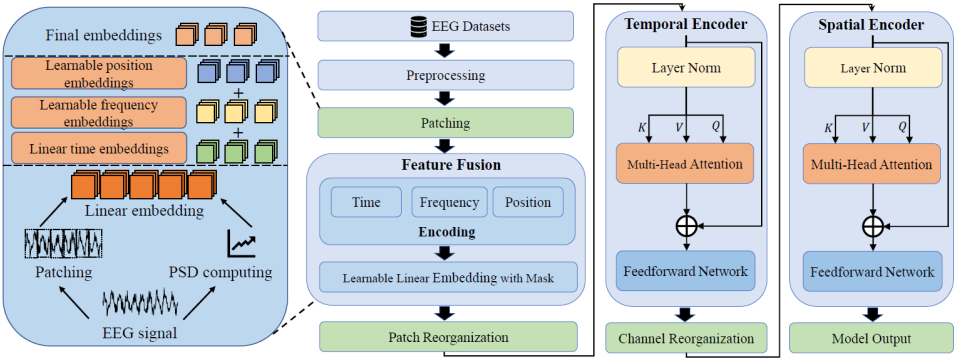


Breaking Down Foundation Models: Fine-tuning

The pre-trained model is then adapted to specific tasks using smaller labeled datasets. This process leverages the knowledge acquired during pre-training to improve performance on the target task.

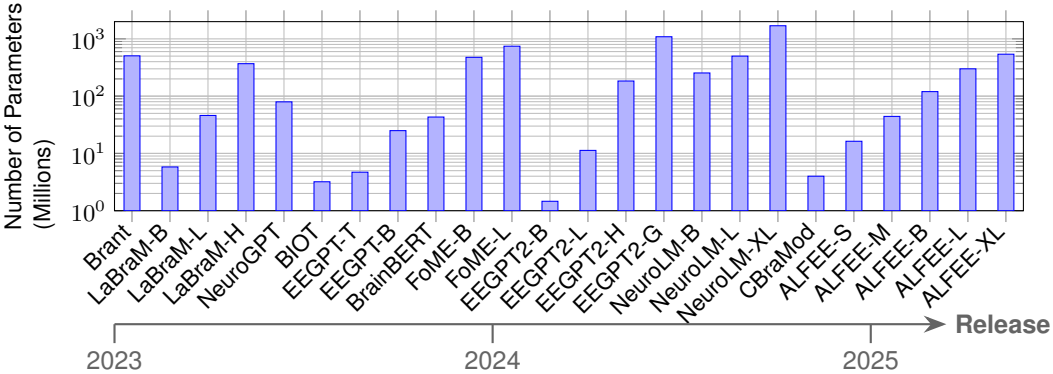


Overview of the architecture and workflow of general EEG-FM Encoder



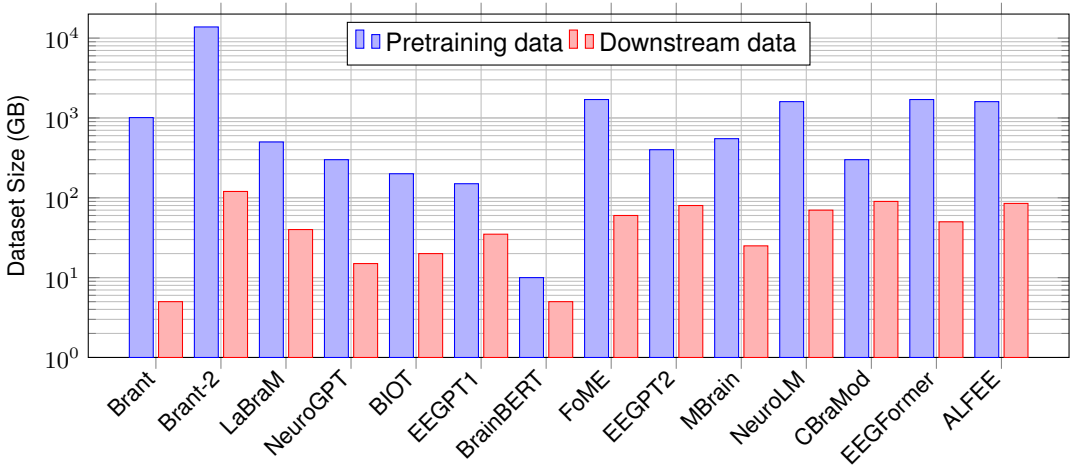
Source: [Lai et al., 2025]

Model Sizes of EEG Foundation Models

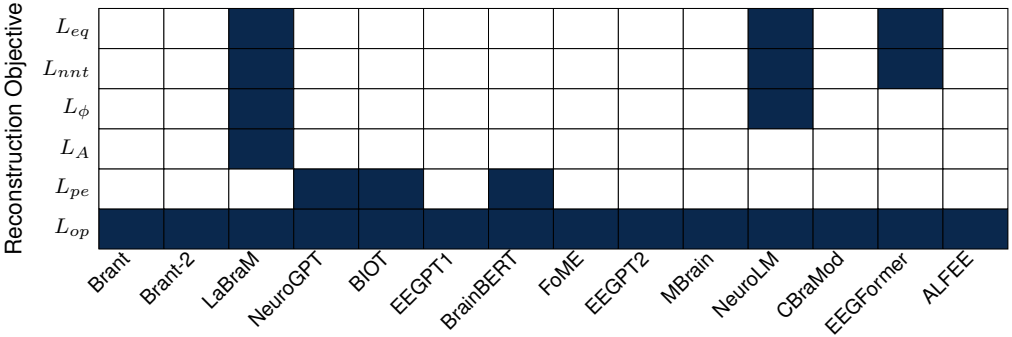


For comparison, GPT-3 has 175,000 million parameters, and GPT-4 is estimated to have 1.5 trillion parameters.

Pretraining vs Downstream Data Scale



Binary Heatmap of Pretraining Objectives



Reconstruction tasks used during pretraining for different models: $\boldsymbol{L_{op}}$, patch reconstruction; $\boldsymbol{L_{pe}}$, embedding reconstruction; $\boldsymbol{L_A}/\boldsymbol{L_\phi}$, Fourier amplitude/phase regression; $\boldsymbol{L_{nnt}}/\boldsymbol{L_{eq}}$, neural-tokenizer codebook/commitment prediction.

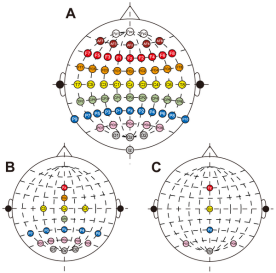
Downstream Task Performance Summary

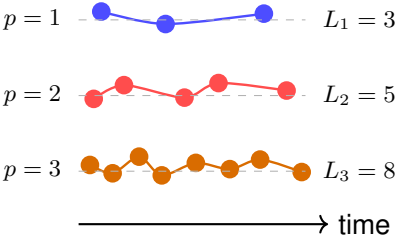
Model	Task	Dataset	Accuracy (%)
LaBraM	Abnormal detection	TUAB	83.8
BENDR	Epilepsy detection	Epilepsy	74.0
NeuroGPT	Epilepsy detection	Epilepsy	73.4
LaBraM	OCD classification	OCD	74.0
LaBraM	Stress classification	Real-world	90.5
CBraMod	Mental workload	EEGMAT	88.9
LaBraM	Mental workload	EEGMAT	85.8
CBraMod	Sleep staging	SHHS	73.5
BIOT	Sleep staging	SHHS	72.2

[Kastrati et al., 2025, Wu et al., 2025, Lai et al., 2025]

Preprocessing & Normalization Challenges

- Most EEG foundation models rely on minimal or inconsistent preprocessing
- Impact of noise, artifacts, and outliers is poorly understood
- Strong variability across datasets, devices, and acquisition protocols



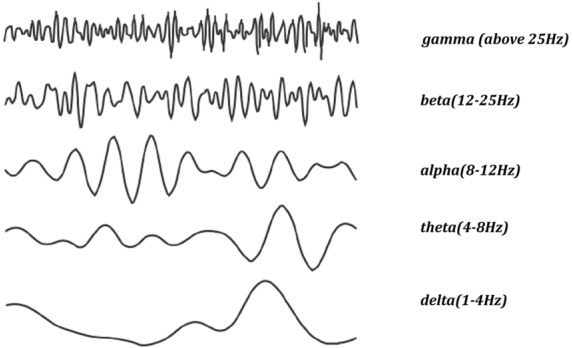


Evaluation & Benchmarking Limitations

- No standardized benchmarks for EEG foundation models
- Evaluation often limited to in-distribution settings
- Few studies report few-shot or zero-shot transfer
- Cross-dataset generalization remains underexplored

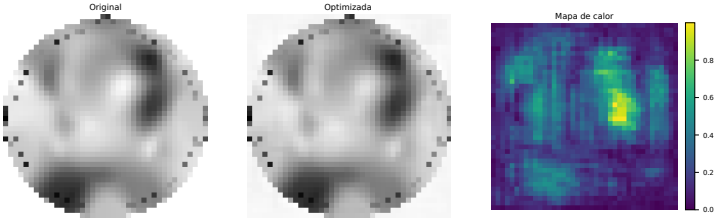
Long-Term Context Modeling

- EEG exhibits multiple temporal scales
- Millisecond-level events (e.g., ERPs) vs hour-long dynamics (e.g., sleep)
- Current architectures struggle to unify short- and long-range context



Interpretability

- Limited focus on model interpretability
- Black-box behavior hinders clinical adoption
- Difficult to relate learned representations to neurophysiology
- Trust is critical for high-stakes diagnostic applications



We gratefully acknowledge the support of

This work has been possible thanks to the collaboration of these institutions.

[Kastrati et al., 2025] Kastrati, A., Bürki, J., Lauer, J., Xuan, C., Iaquinto, R., and Wattenhofer, R. (2025).
Eeg-bench: A benchmark for eeg foundation models in clinical applications.

[Lai et al., 2025] Lai, J., Wei, J., Yao, L., and Wang, Y. (2025).
A simple review of eeg foundation models: Datasets, advancements and future perspectives.

[Wu et al., 2025] Wu, J., Ren, Z., Wang, J., Zhu, P., Song, Y., Liu, M., Zheng, Q., Bai, L., Ouyang, W., and Song, C. (2025).
Adabrain-bench: Benchmarking brain foundation models for brain-computer interface applications.
arXiv preprint arXiv:2507.09882.
Cross-subject: CBraMod 88.9%, LaBraM 85.8% on EEGMAT. Multi-subject: LaBraM 70.9%, CBraMod 70.3% on SEED.